Quality of Service (QoS) Concepts

6CS029 Advanced Networking

In this session

- Network Transmission Quality
- Traffic Characteristics
- Queuing Algorithms
- QoS Models
- QoS Implementation Techniques

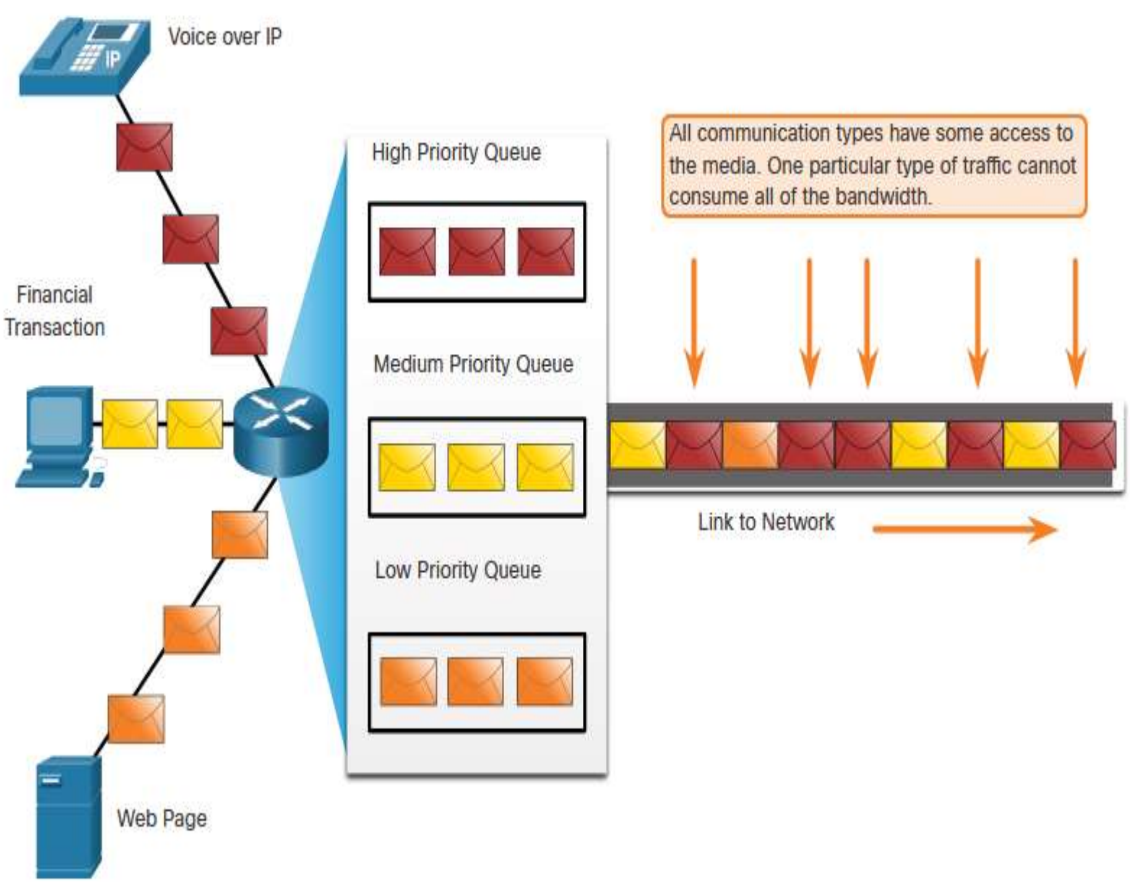
Network Transmission Quality

The Purpose of QoS

- When traffic volume is greater than what can be transported across the network, devices queue (hold) the packets in memory until resources become available to transmit them.
- Queuing packets causes delay because new packets cannot be transmitted until previous packets have been processed.
- If the number of packets to be queued continues to increase, the memory within the device fills up and packets are dropped.

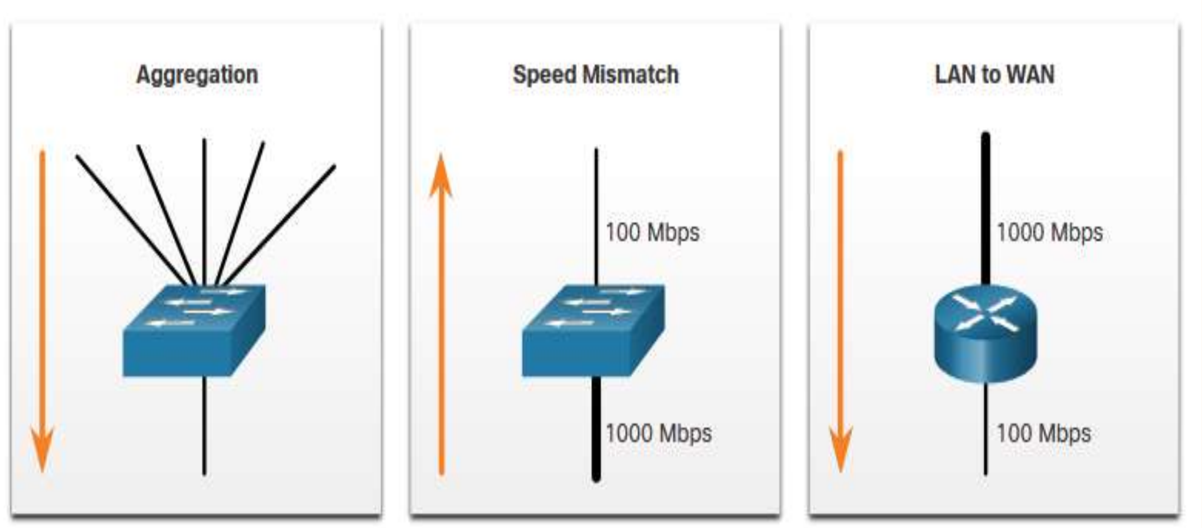
Prioritizing Traffic

- One QoS technique that can help with this problem is to classify data into multiple queues



Bandwidth, Congestion, Delay, and Jitter

- Bandwidth is measured in the number of bits that can be transmitted in a single second.
- Network congestion causes delay.
- Congestion points are ideal candidates for QoS mechanisms.
  - Aggregation, speed mismatch, and LAN to WAN.



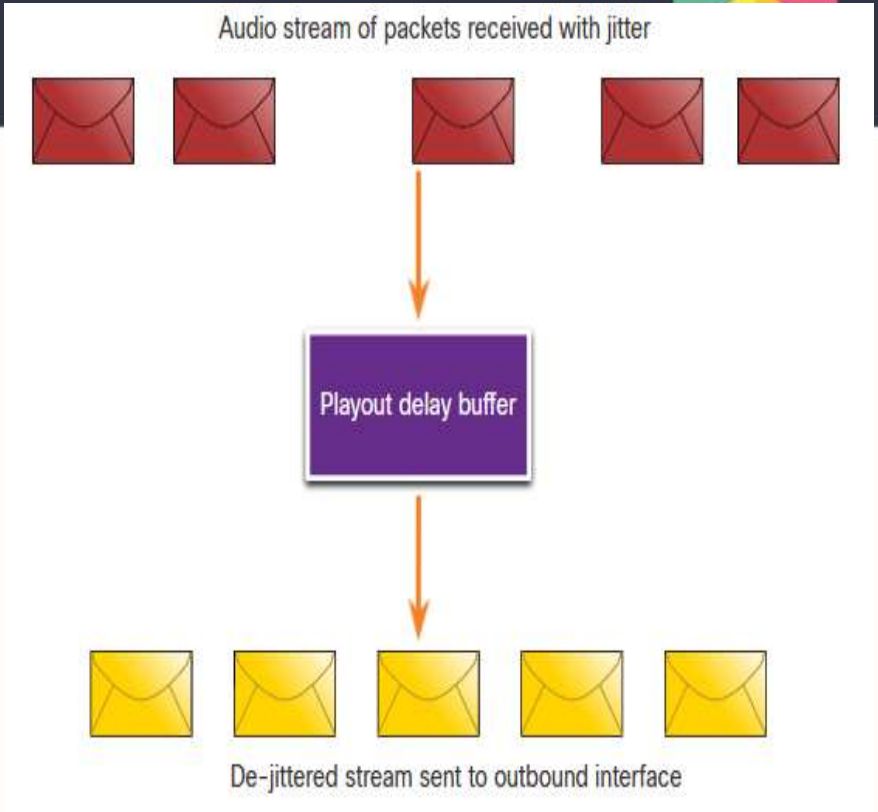
Bandwidth, Congestion, Delay, and Jitter

Delay	Description
Code delay	The fixed amount of time it takes to compress data at the source before transmitting to the first internetworking device, usually a switch.
Packetization delay	The fixed time it takes to encapsulate a packet with all the necessary header information.
Queuing delay	The variable amount of time a frame or packet waits to be transmitted on the link.
Serialization delay	The fixed amount of time it takes to transmit a frame onto the wire.
Propagation delay	The variable amount of time it takes for the frame to travel between the source and destination.
De-jitter delay	The fixed amount of time it takes to buffer a flow of packets and then send them out in evenly spaced intervals.

Packet Loss

Without QoS mechanisms, time-sensitive packets, such as real-time video and voice, are dropped with the same frequency as data that is not time-sensitive.

- When a router receives a Real-Time Protocol (RTP) digital audio stream for Voice over IP (VoIP), it compensates for the jitter that is encountered using a playout delay buffer.
- The playout delay buffer buffers these packets and then plays them out in a steady stream.

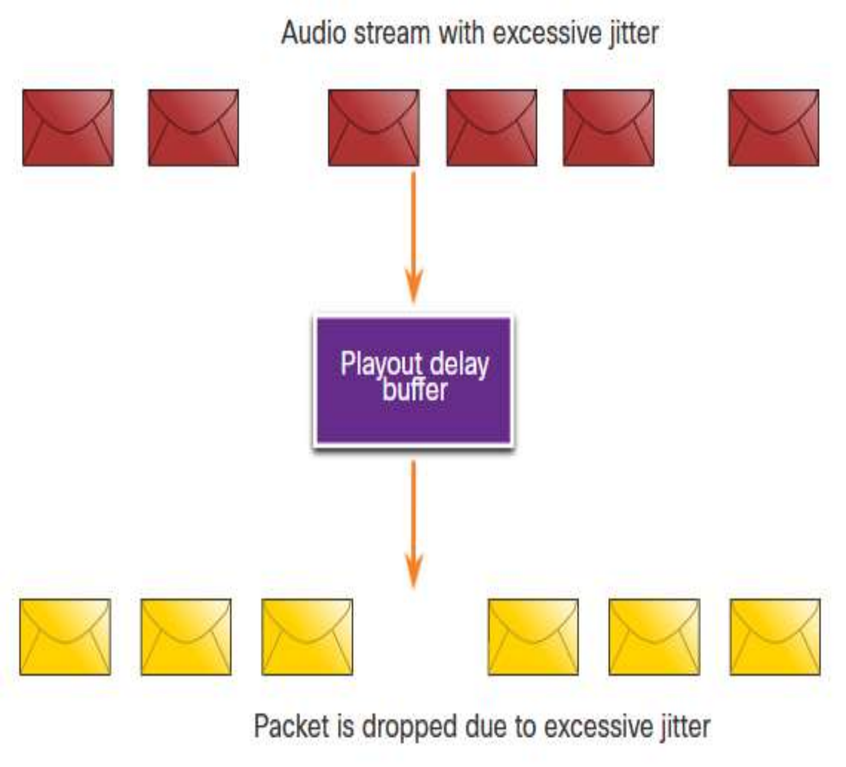


Playout Delay Buffer Compensates for Jitter

Packet Loss

If the jitter is so large that it causes packets to be received out of the range of the play out buffer, the out-of-range packets are discarded and dropouts are heard in the audio.

- For losses as small as one packet, the digital signal processor (DSP) interpolates what it thinks the audio should be and no problem is audible to the user.
- When jitter exceeds what the DSP can do to make up for the missing packets, audio problems are heard.



Traffic Characteristics

Network Traffic Trends

In the early 2000s, the predominant types of IP traffic were voice and data.

- Voice traffic has a predictable bandwidth need and known packet arrival times.
- Data traffic is not real-time and has unpredictable bandwidth need.
- Data traffic can temporarily burst

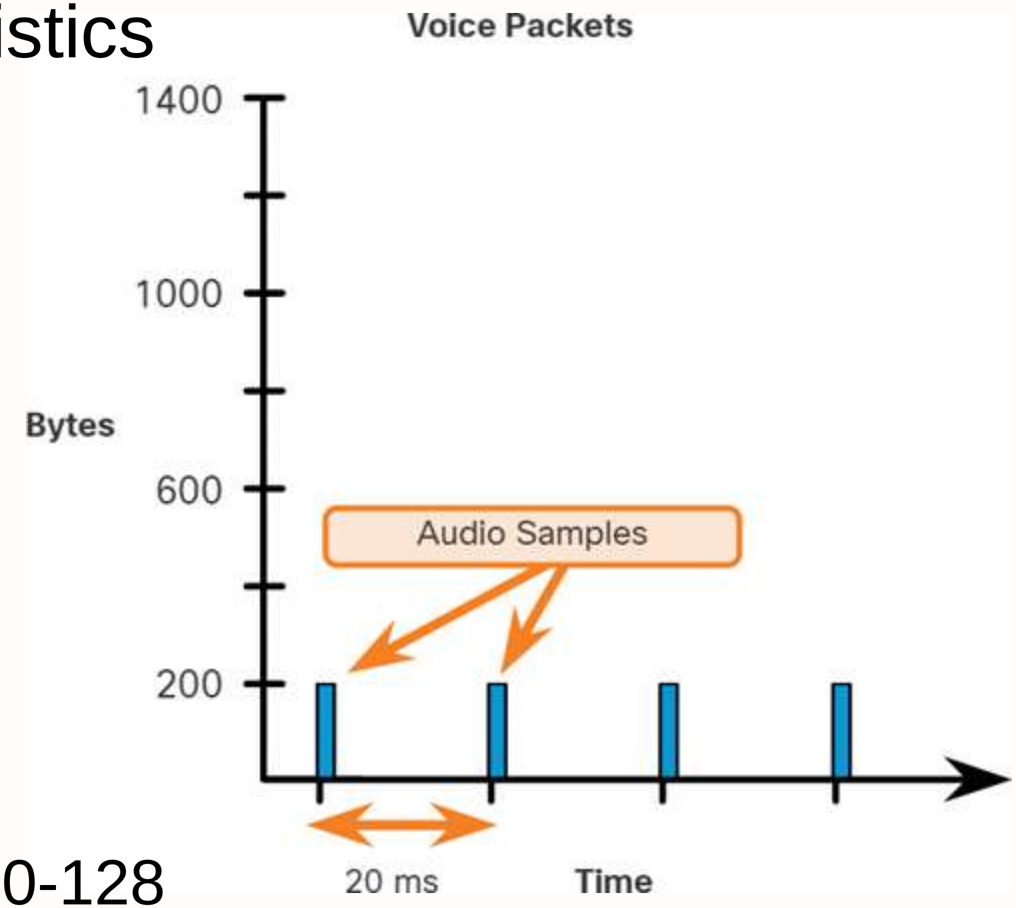
More recently, video traffic has become the increasingly important.

- Video traffic represented 70% of all traffic in 2017.
- By 2022, video will represent 82% of all traffic.
- Mobile video traffic will reach 60.9 exabytes per month by 2022.

The type of demands that voice, video, and data traffic place on the network are very different.

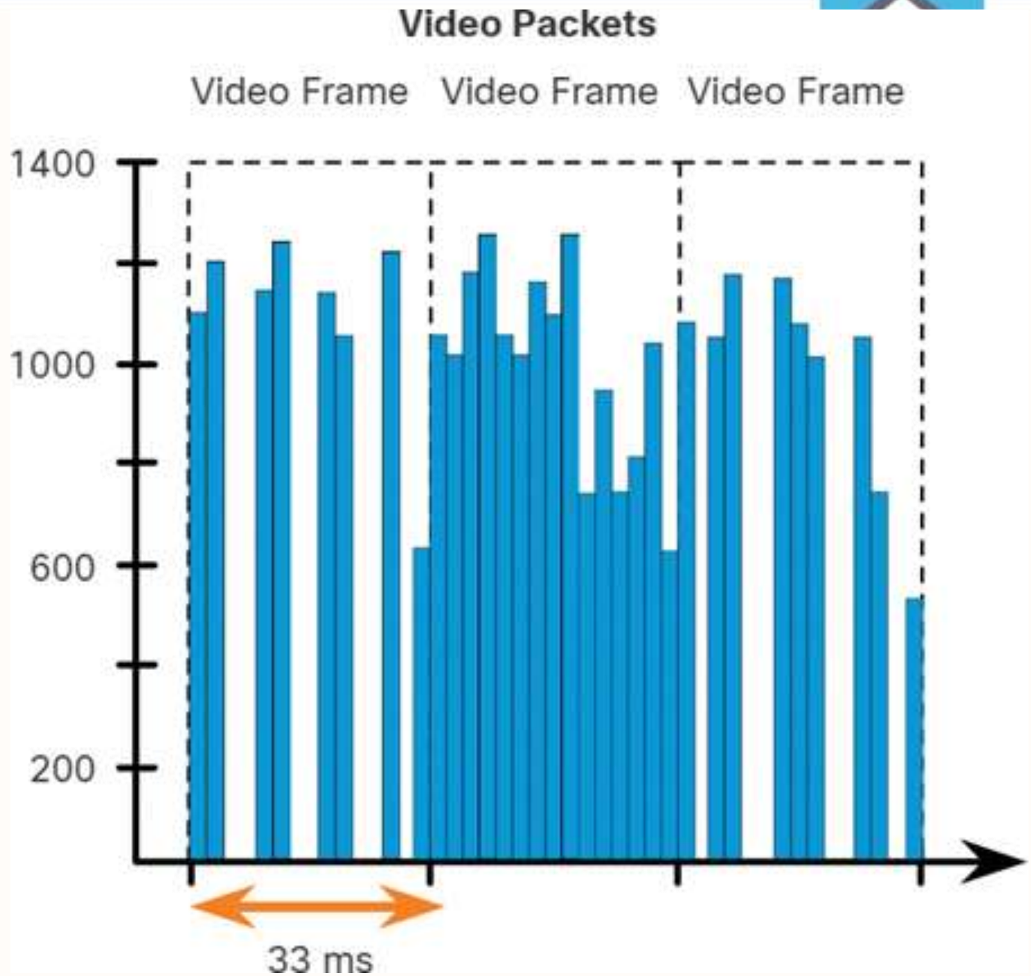
Voice

- Voice Traffic Characteristics
  - Smooth
  - Benign
  - Drop sensitive
  - Delay sensitive
  - UPD priority
- Requirements
  - Latency < 150ms
  - Jitter < 30ms
  - Loss < 1% Bandwidth (30-128 Kbps)



Video

- Characteristics
  - Bursty
  - Greedy
  - Drop sensitive
  - Delay sensitive
  - UPD priority
- One-Way Requirements
  - Latency ≤ 200-400 ms
  - Jitter ≤ 30-50 ms
  - Loss ≤ 0.1 – 1%
  - Bandwidth (384 Kbps - 20 Mbps)



Data

- Data applications that have no tolerance for data loss, such as email and web pages, use TCP to ensure that if packets are lost in transit, they will be resent.
  - Data traffic can be smooth or bursty.
  - Network control traffic is usually smooth and predictable.
- Some TCP applications can consume a large portion of network capacity.

- Characteristics
  - Smooth/bursty
  - Benign/greedy
  - Drop insensitive
  - Delay insensitive
  - TCP retransmits

Data

Data traffic is relatively insensitive to drops and delays compared to voice and video. Quality of Experience or QoE is important to consider with data traffic.

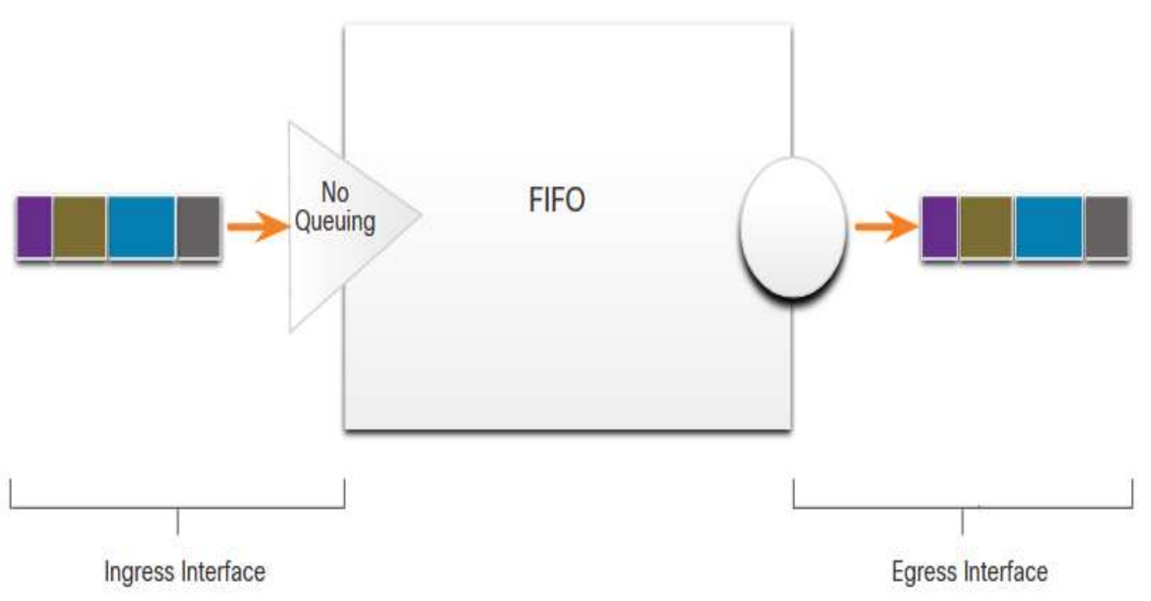
- Does the data come from an interactive application?
- Is the data mission critical?

Factor	Mission Critical	Not Mission Critical
Interactive	Prioritize for the lowest delay of all data traffic and strive for a 1 to 2 second response time.	Applications could benefit from lower delay.
Not interactive	Delay can vary greatly as long as the necessary minimum bandwidth is supplied.	Gets any leftover bandwidth after all voice, video, and other data application needs are met.

Queuing Algorithms

First in First Out

- First In First Out (FIFO) queuing buffers and forwards packets in the order of their arrival.
- FIFO has no concept of priority or classes of traffic and consequently, makes no decision about packet priority.
- There is only one queue, and all packets are treated equally.
- Packets are sent out an interface in the order in which they arrive.

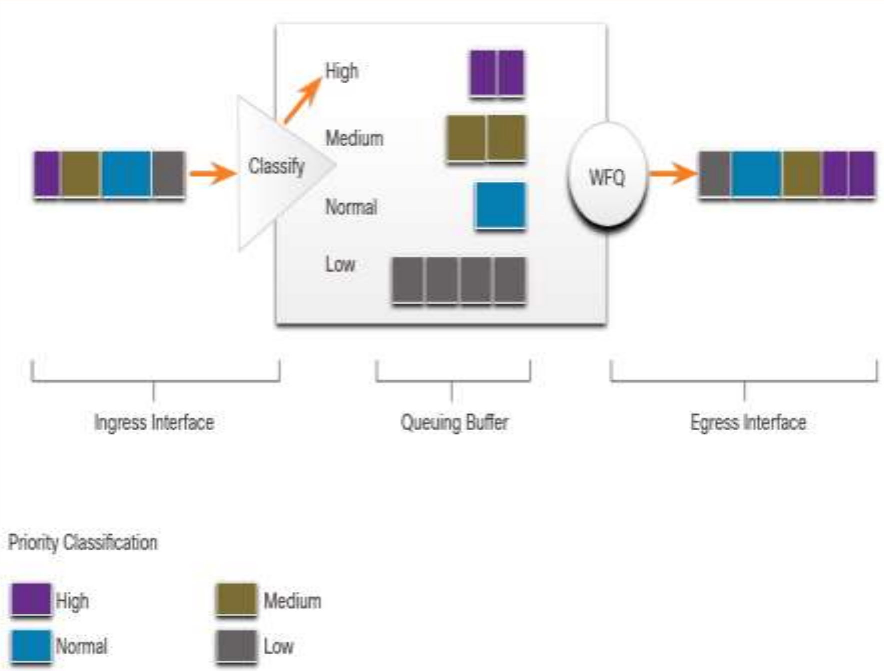


Weighted Fair Queuing (WFQ)

Provides fair bandwidth allocation to all network traffic.

- WFQ applies priority to identified traffic, classifies it into conversations or flows, and then determines how much bandwidth each flow is allowed relative to other flows.
- Classification can be based on source and destination IP addresses, MAC addresses, port numbers, protocol, and Type of Service (ToS) value.

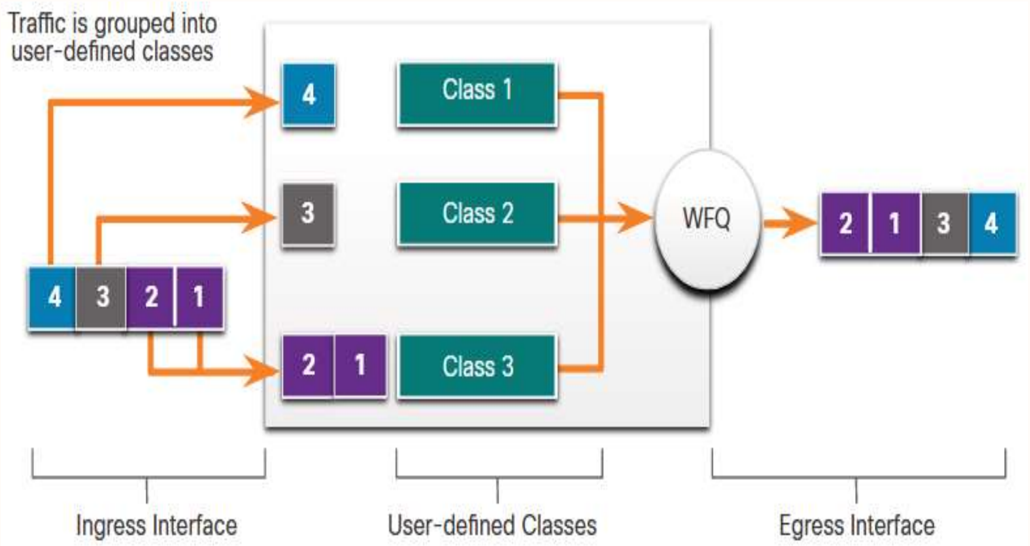
- WFQ is not supported with tunneling and encryption



Class-Based Weighted Fair Queuing (CBWFQ)

Extends the standard WFQ functionality to provide support for user-defined traffic classes.

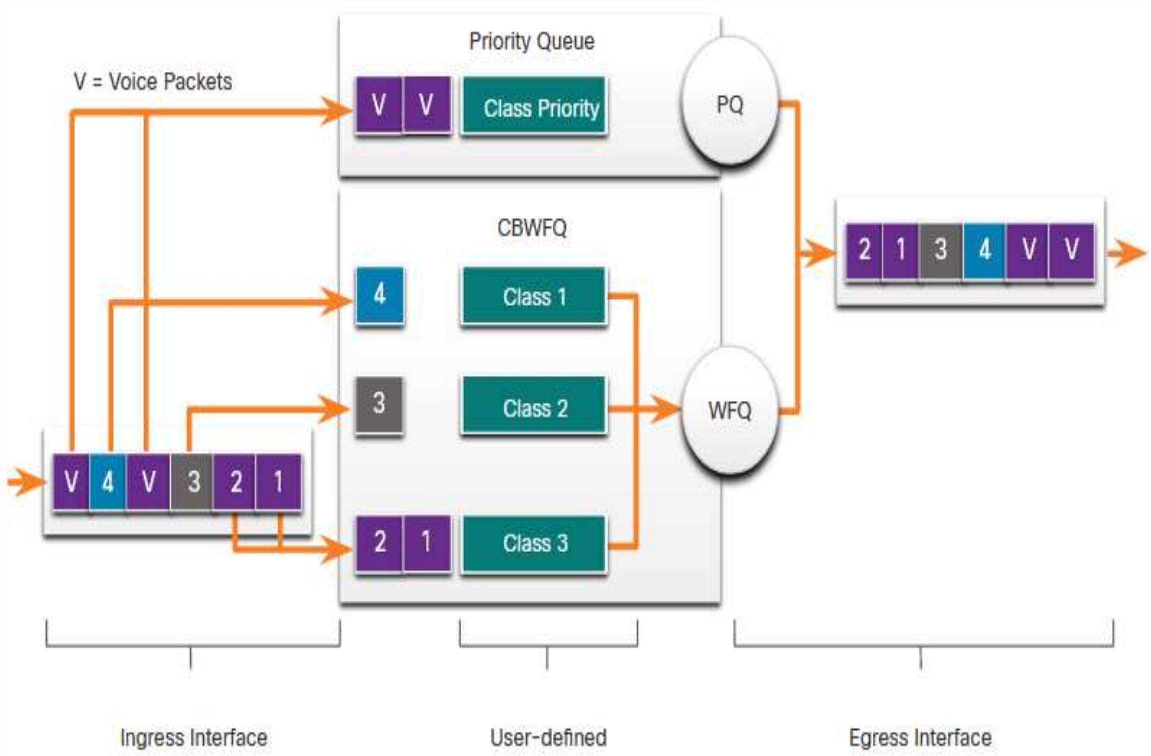
- Traffic classes are defined based on match criteria including protocols, access control lists (ACLs), and input interfaces.
- A FIFO queue is reserved for each class
- A class can be assigned characteristics, such as bandwidth, weight, and maximum packet limit.



Low Latency Queuing (LLQ)

The Low Latency Queuing (LLQ) feature brings strict priority queuing (PQ) to CBWFQ.

- Strict PQ allows delay-sensitive packets such as voice to be sent before packets in other queues.
- LLQ allows delay-sensitive packets to be sent first giving delay-sensitive packets preferential treatment over other traffic.



QoS Models

QoS Models

Model	Description
Best-effort model	• Not an implementation as QoS is not explicitly configured. • Use when QoS is not required.
Integrated services (IntServ)	• Provides very high QoS to IP packets with guaranteed delivery. • Defines a signaling process for applications to signal to the network that they require special QoS for a period and that bandwidth should be reserved. • IntServ can severely limit the scalability of a network.
Differentiated services (DiffServ)	• Provides high scalability and flexibility in implementing QoS. • Network devices recognize traffic classes and provide different levels of QoS to different traffic classes.

Best Effort

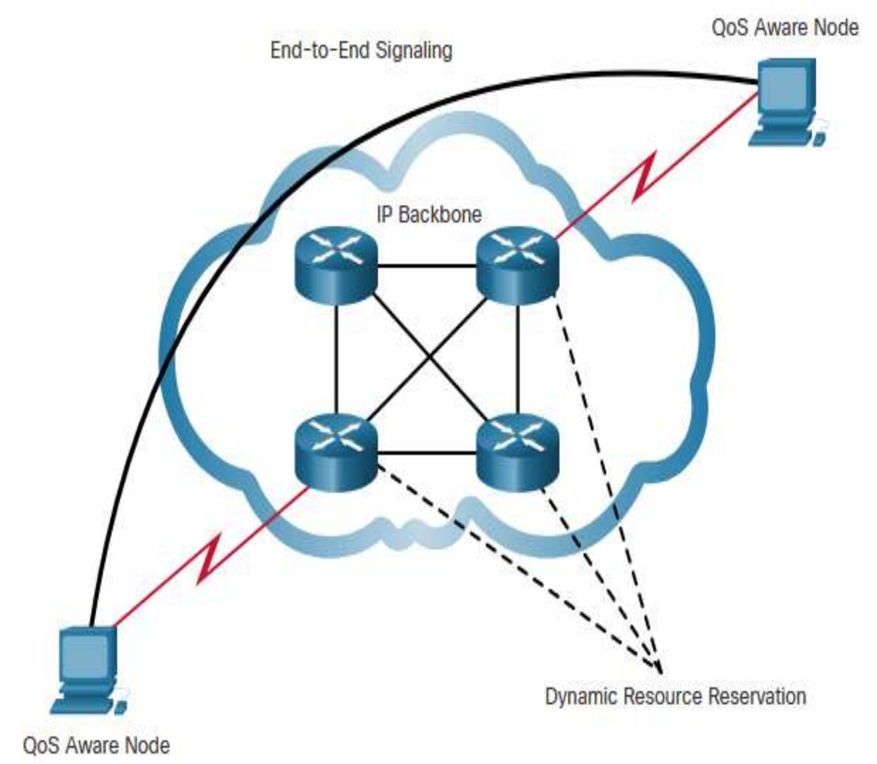
The basic design of the internet is best-effort packet delivery and provides no guarantees.

- All network packets are treated in the same way

Benefits	Drawbacks
The model is the most scalable.	There are no guarantees of delivery.
Scalability is only limited by available bandwidth, in which case all traffic is equally affected.	Packets will arrive whenever they can and in any order possible, if they arrive at all.
No special QoS mechanisms are required.	No packets have preferential treatment.
It is the easiest and quickest model to deploy.	Critical data is treated the same as casual email is treated.

Integrated Services

- Explicitly provide QoS to individual flows
- Uses resource reservation and admission-control mechanisms
- Uses a connection-oriented approach
- Admission control at the edge router
- Application informs the network of its traffic profile and requests a particular kind of service.
- Uses the Resource Reservation Protocol (RSVP) to signal the QoS needs of an application's traffic along devices in the end-to-end path through the network.



Benefits

- Explicit end-to-end resource admission control
- Per-request policy admission control
- Signaling of dynamic port numbers

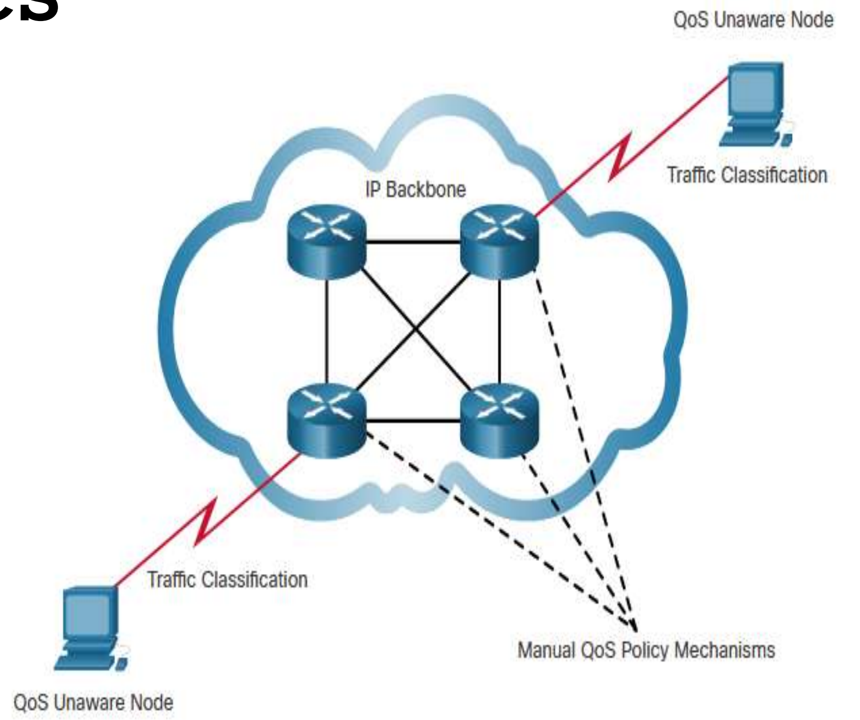
Drawbacks

- Resource intensive due to the stateful architecture requirement for continuous signaling.
- Flow-based approach not scalable to large implementations such as the internet.

Differentiated Services

Simple and scalable mechanism for classifying and managing network traffic.

- Is not an end-to-end QoS strategy because it cannot enforce end-to-end guarantees.
- The router which classifies the flows into classes based on business requirements and provides the appropriate QoS policy for the classes.
- Enforces and applies QoS mechanisms on a hop-by-hop basis
- It is possible to choose many levels of service with DiffServ.



Benefits

- Highly scalable
- Provides many different levels of quality

Drawbacks

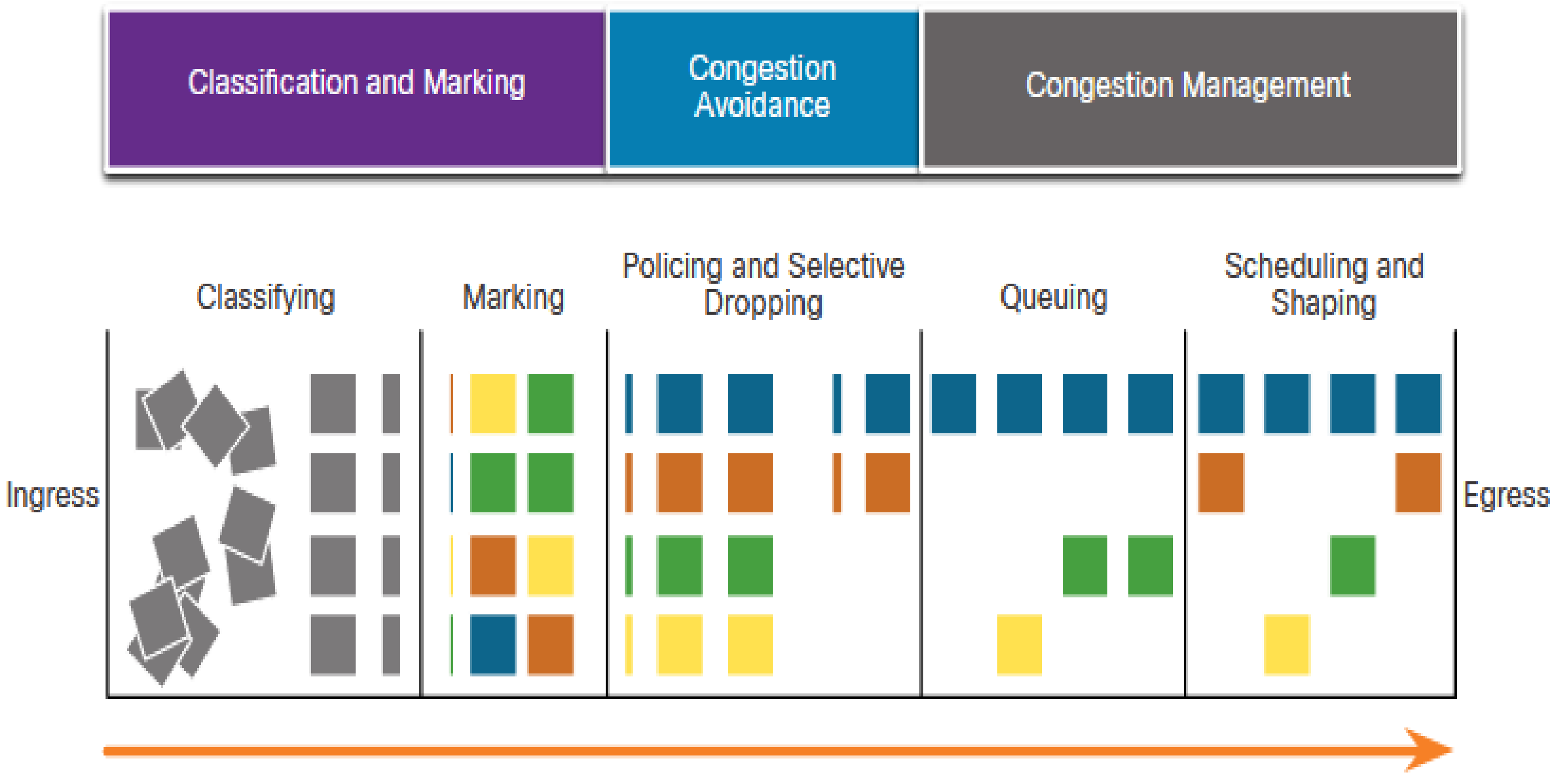
- No absolute guarantee of service quality
- Requires a set of complex mechanisms to work in concert throughout the network

QoS Implementation Techniques

QoS Tools

QoS Tools	Description
Classification and marking tools	• Sessions, or flows, are analyzed to determine what traffic class they belong to and then marked
Congestion avoidance tools	• Traffic classes are allotted portions of network resources, as defined by the QoS policy. • The QoS policy also identifies how some traffic may be selectively dropped, delayed, or re-marked to avoid congestion.
Congestion management tools	• When traffic exceeds available network resources, traffic is queued to await availability of resources.

QoS Tools



Note: Classification and marking can be done on ingress or egress, whereas other QoS actions such queuing and shaping are usually done on egress.

Classification and Marking

Before a packet can have a QoS policy applied to it, the packet has to be classified.

Classification determines the class of traffic to which packets or frames belong. Only after traffic is marked can policies be applied to it.

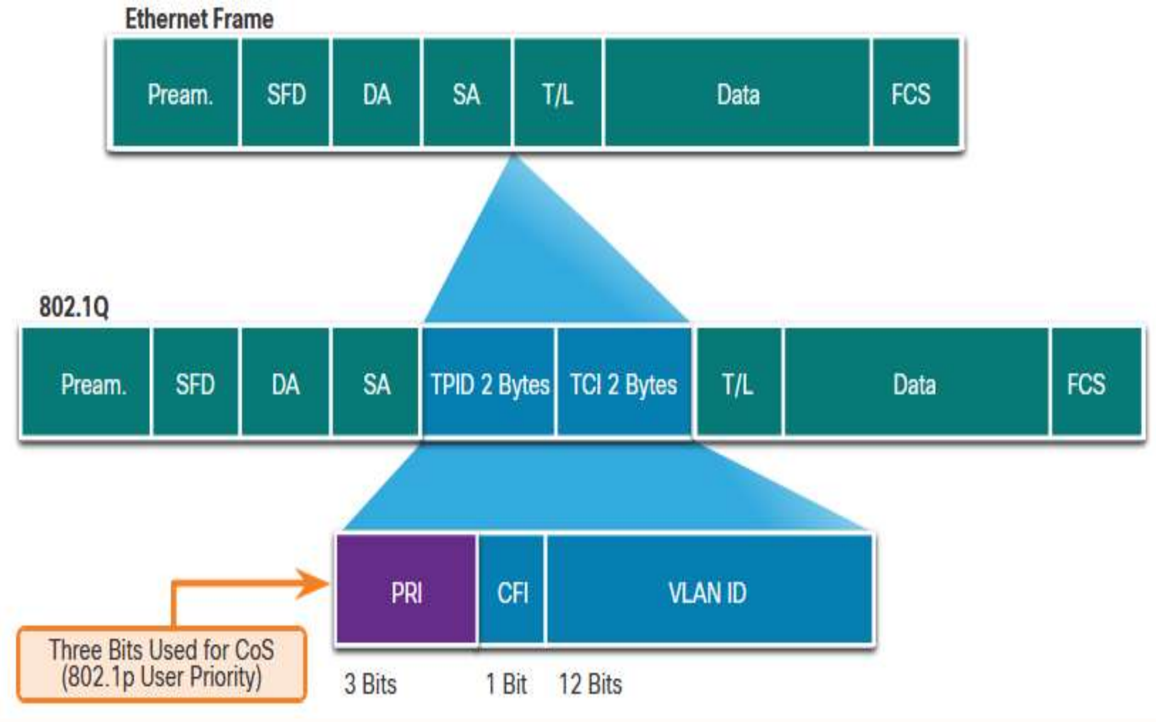
How a packet is classified depends on the QoS implementation.

- Methods of classifying traffic flows at Layer 2 and 3 include using interfaces, ACLs, and class maps.
- Traffic can also be classified at Layers 4 to 7 using **Network Based Application Recognition (NBAR).**

QoS Tools	Layer	Marking Field	Bits
Ethernet (802.1q, 802.1p)	2	Class of Service (CoS)	3
802.11 (Wi-Fi)	2	Wi-Fi Traffic Identifier (TID)	3
MPLS	2	Experimental (EXP)	3
IPv4 and IPv6	3	IP Precedence (IPP)	3
IPv4 and IPv6	3	Differentiated Services Code Point (DSCP)	6

Marking at Layer 2

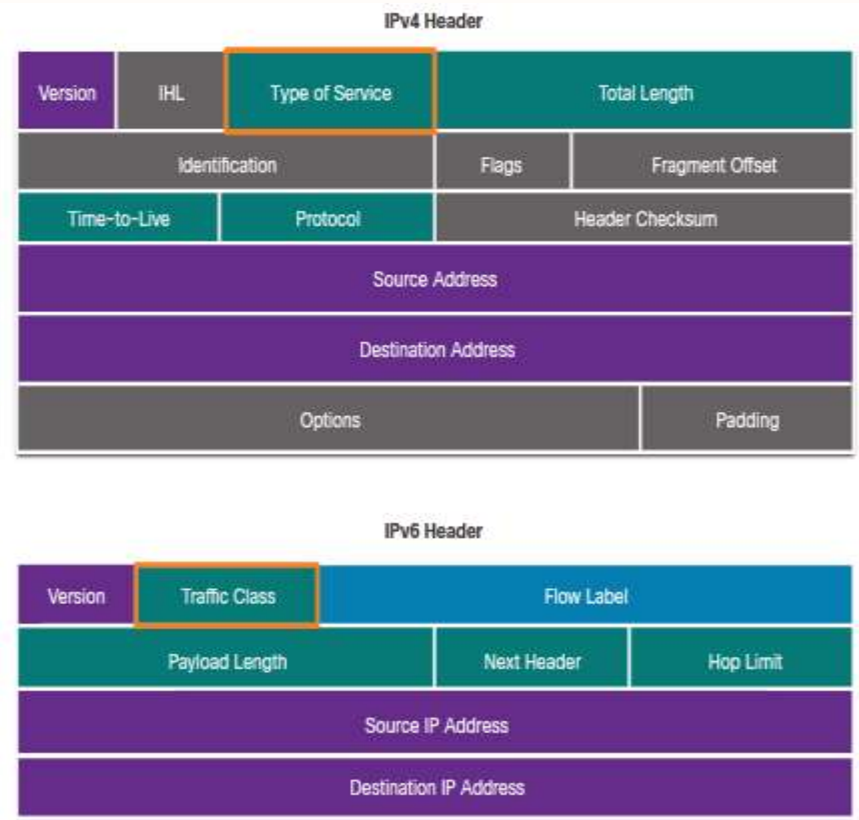
802.1Q is the IEEE standard that supports VLAN tagging at Layer 2 on Ethernet networks. When 802.1Q is implemented, two fields are inserted into the Ethernet frame following the source MAC address field.



Marking at Layer 3

IPv4 and IPv6 specify an 8-bit field in their packet headers to mark packets.

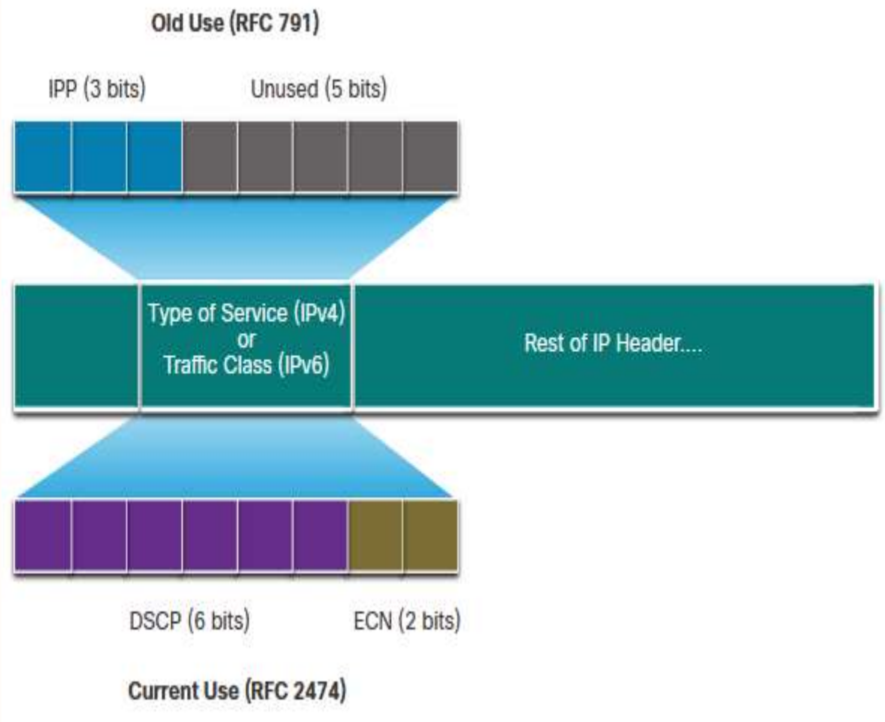
Both IPv4 and IPv6 support an 8-bit field for marking: the Type of Service (ToS) field for IPv4 and the Traffic Class field for IPv6.



Type of Service and Traffic Class Field

The Type of Service (IPv4) and Traffic Class (IPv6) carry the packet marking as assigned by the QoS classification tools.

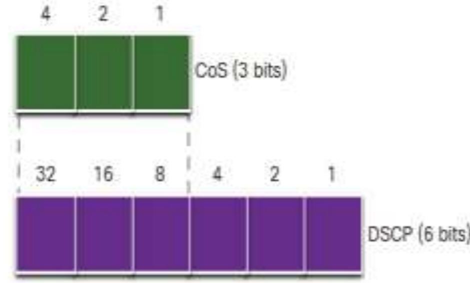
- RFC 791 specified the 3-bit IP Precedence (IPP) field to be used for QoS markings.
- RFC 791 and redefines the ToS field by renaming and extending the IPP field to 6 bits called the Differentiated Services Code Point (DSCP) field, these six bits offer a maximum of 64 possible classes of service.
- The remaining two IP Extended Congestion Notification (ECN) bits can be used by ECN-aware routers to mark packets instead of dropping them.



DSCP Values

The 64 DSCP values are organized into three categories:

- **Best-Effort (BE)** - This is the default for all IP packets. The DSCP value is 0.
- **Expedited Forwarding (EF)** - decimal value 46 (binary **101**110). The first 3 bits (101) map directly to the Layer 2 CoS value 5 used for voice traffic. At Layer 3, it is recommended that EF only be used to mark voice packets.
- **Assured Forwarding (AF)** - use the 5 most significant DSCP bits to indicate queues and drop preference.



CoS values, Class Selectors, and corresponding DSCP 6-bit value

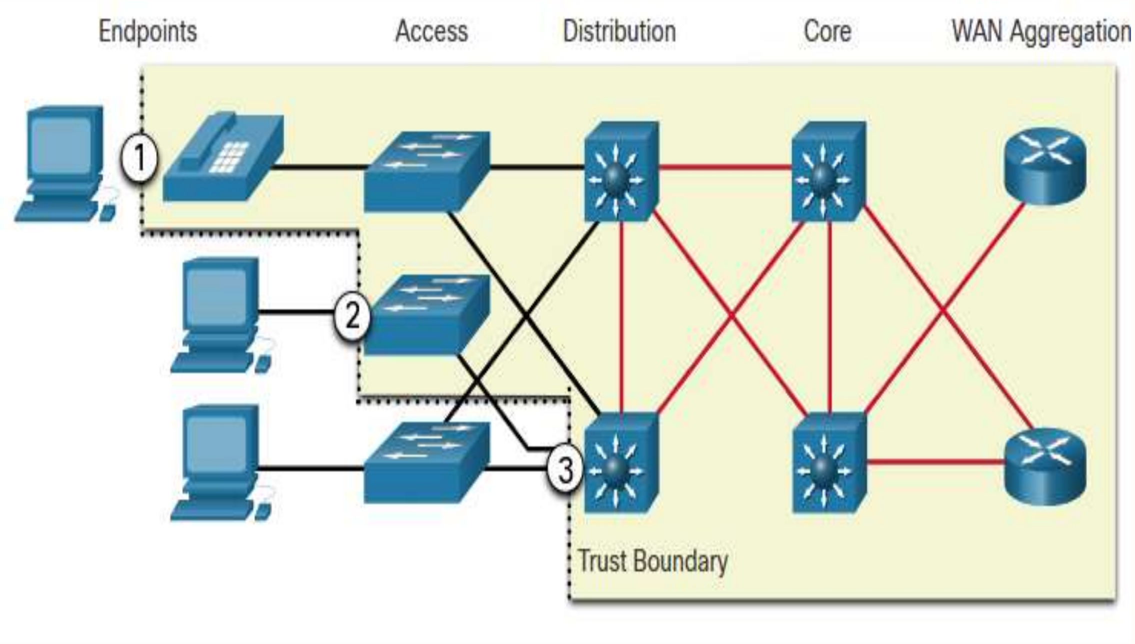
CoS Value	CoS Binary Value	Class Selector (CS)	CS Binary	DSCP Decimal Value
0	000	CS0*/DF	000 000	0
1	001	CS1	001 000	8
2	010	CS2	010 000	16
3	011	CS3	011 000	24
4	100	CS4	100 000	32
5	101	CS5	101 000	40
6	110	CS6	110 000	48
7	111	CS7	111 000	56

- The first 3 most significant bits of the DSCP field and indicate the class.
- Map directly to the 3 bits of the CoS field and the IPP field to maintain compatibility

Trust Boundaries

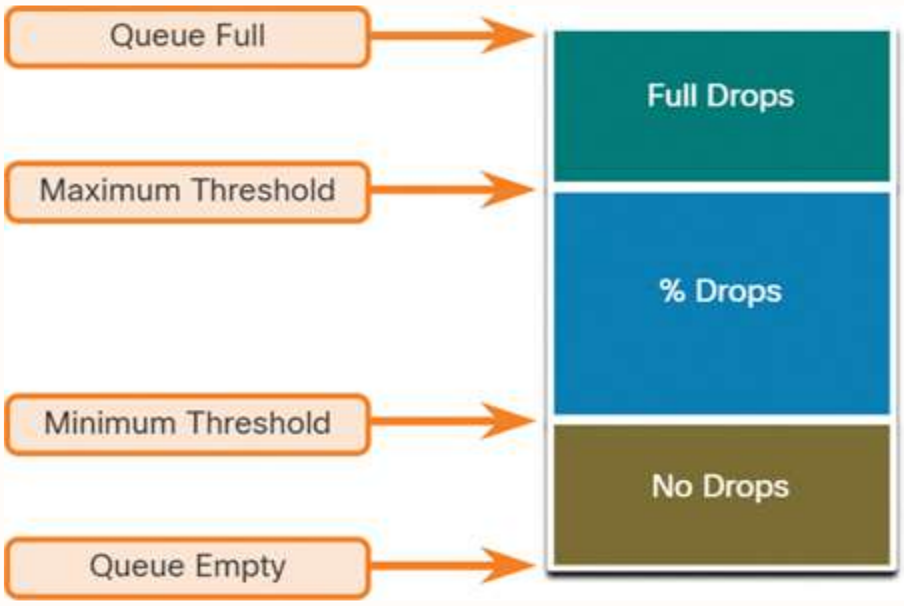
Traffic should be classified and marked as close to its source as technically and administratively feasible. This defines the trust boundary.

1. Trusted endpoints have the capabilities and intelligence to mark application traffic to the appropriate Layer 2 CoS and/or Layer 3 DSCP values.
2. Secure endpoints can have traffic marked at the Layer 2 switch.
3. Traffic can also be marked at Layer 3 switches / routers.



Congestion Avoidance

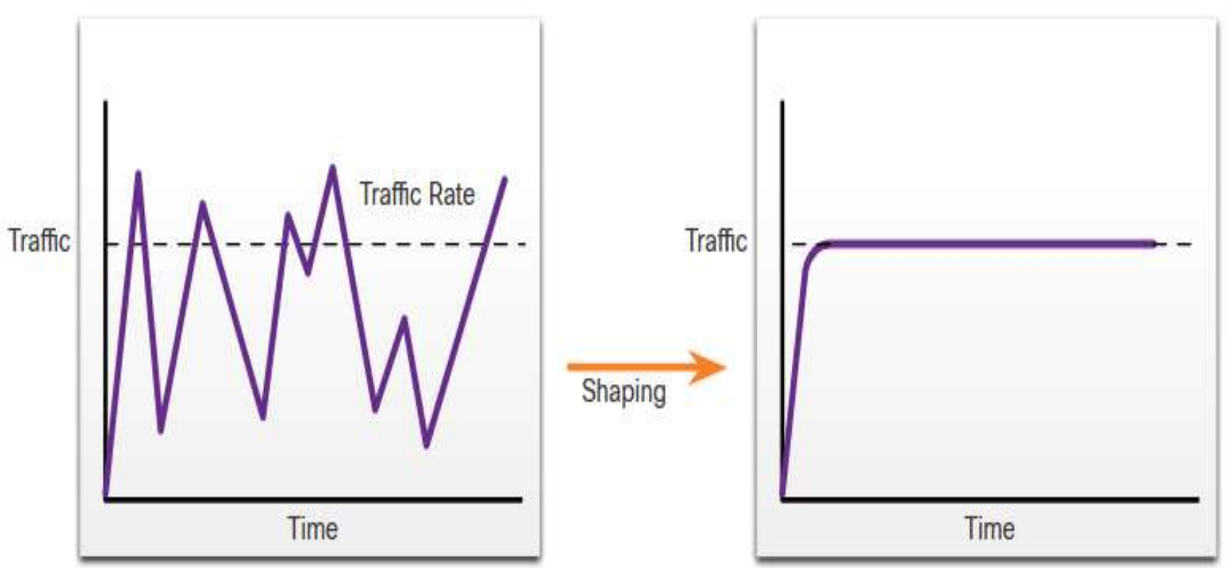
- Tools to monitor traffic to anticipate and avoid congestion at common bottlenecks
- monitor the average depth of the queue, When the maximum threshold is passed, all packets are dropped.
- Some congestion avoidance techniques provide preferential treatment for which packets get dropped.



Shaping and Policing

Mechanisms provided to prevent congestion.

- Traffic shaping retains excess packets in a queue and then schedules the excess for later transmission over increments of time. Traffic shaping results in a smoothed packet output rate.
- Shaping is an outbound concept; packets going out an interface get queued and can be shaped. In contrast, policing is applied to inbound traffic on an interface.



Shaping and Policing

Policing is applied to inbound traffic on an interface. Policing is commonly implemented by service providers to enforce a contracted customer information rate (CIR). However, the service provider may also allow bursting over the CIR if the service provider's network is not currently experiencing congestion.

Questions?